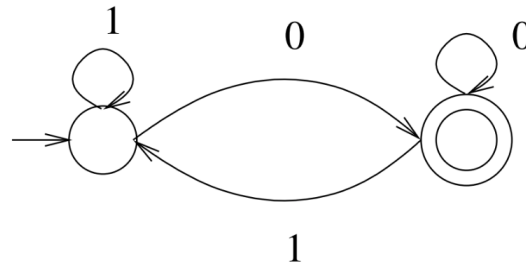


Question 1

The following questions are multiple choice. There is at least one correct choice but there may be several. To get all the marks you have to list all the correct answers and none of the wrong ones.

1. Given the following finite automaton A1 over $\Sigma = \{0, 1\}$.



Which of the following statements about A1 are correct?

True False

A1 is a Nondeterministic Finite Automaton (NFA) but not a Deterministic Finite Automaton (DFA).

☐ True ☐ False

A1 is a Deterministic Finite Automaton (DFA).

☐ True ☐ False

$10 \in L(A1)$

☐ True ☐ False

A1 accepts all words which end with 0 (but possibly more).

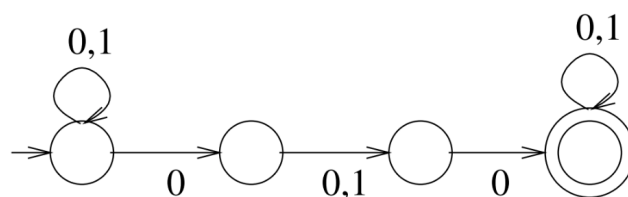
☐ True ☐ False

A1 accepts all words which contain a 0 (but possibly more).

☐ True ☐ False

(5 marks)

2. Given the following finite automaton A2 over $\Sigma = \{0, 1\}$.



Which of the following statements about A₂ are correct?

True False

A2 is a Nondeterministic Finite Automaton (NFA) but not a Deterministic Finite Automaton (DFA).

☐ True ☐ False

A2 is a Deterministic Finite Automaton (DFA).

☐ True ☐ False

$0101 \in L(A2)$

☐ True ☐ False

A2 accepts all words containing 010 (but possibly more).

☐ True ☐ False

A2 accepts all words whose penultimate symbol (the one before the last) is 1 (but possibly more).

☐ True ☐ False

(5 marks)

3. Which of the following statements are correct?

True False

A language is a sequence of words.

☐ ☐

A word is a sequence of symbols

☐ ☐

Deterministic and Nondeterministic Finite Automata recognize the same set of languages.

☐ ☐

Deterministic and Nondeterministic Pushdown Automata recognize the same set of languages.

☐ ☐

Deterministic and Nondeterministic Turing Machines recognize the same set of languages.

☐ ☐

(5 marks)

4. Given the following Context-Free Grammar (CFG) with Non-Terminal symbols S, T over $\Sigma = \{ (,), a, b \}$, S the start symbol with productions:

$$S \rightarrow (T) \mid a$$

$$T \rightarrow (S) \mid b$$

Which of the following words are in the language of G (i.e. are elements of $L(G)$)?

True False

(a)

☐ ☐

((a))

☐ ☐

(a)((b))

☐ ☐

(())

☐ ☐

((((a)))

☐ ☐

(5 marks)

5. Which of the following statements about the Halting Problem are correct.

True False

The Halting Problem is semi-decidable (recursively enumerable).

☐ ☐

The negation of the Halting Problem is semi-decidable (recursively enumerable).

☐ ☐

The Halting Problem is the problem that Turing Machines can get stuck.

☐ ☐

There is no Turing Machine which decides the Halting Problem.

☐ ☐

The Halting problem can be expressed as a language.

☐ ☐

(5 marks)

Question 2

Give Regular Expressions for the following languages over $\Sigma = \{a, b, c\}$:

6. Words that contain no as or bs .

(2 marks)

7. Words that only contain as or bs.

(2 marks)

8. Words that contain an even number of bs.

(2 marks)

9. Words that contain the sequence aa.

(2 marks)

10. Words that do not contain the sequence aa.

(2 marks)

Given the alphabet.

inductive Σ_{ABC} : Type
 $| a | b | c$

-- Consider the following languages over Σ_{ABC}

abbrev L_1 : Lang Σ_{ABC}
-- words that contain the same number of as and bs.
 $:= \{ w : \text{Word } \Sigma_{ABC} \mid w.\text{count } a = w.\text{count } b \}$

abbrev L_2 : Lang Σ_{ABC}
-- words that contain the same number of as and bs modulo 2

$:= \{ w : \text{Word SigmaABC} \mid$
 $w.\text{count } a \equiv w.\text{count } b \pmod{2} \}$

abbrev $L_3 : \text{Lang SigmaABC}$
 -- only the words a, aa, aaa
 $:= \{ [a], [a,a], [a,a,a] \}$

abbrev $L_4 : \text{Lang SigmaABC}$
 -- the complement of L_3 , ie all the words that are not in L_3
 $:= \{ w \mid w \notin L_3 \}$

abbrev $L_5 : \text{Lang SigmaABC}$
 -- the language of repeated words
 $:= \{ w++w \mid (w : \text{Word SigmaABC}) \}$

11. Decide which of the languages are regular.

True False

L_1 is regular. ☐ ☐

L_2 is regular. ☐ ☐

L_3 is regular. ☐ ☐

L_4 is regular. ☐ ☐

L_5 is regular. ☐ ☐

(5 marks)

12. For one of the languages that are regular provide evidence that the language is regular. Clearly state which language you have chosen.

(5 marks)

13. For one of the languages that are not regular provide evidence that the language is not regular. Clearly state which language you have chosen.

(5 marks)

Question 3

We introduce a grammar for regular expressions over the following alphabet

inductive Sigma_RE : Type

| a | b | c

| epsilon -- ε

| empty -- $\emptyset$

| dot -- $\cdot$

| plus -- $+$

| star -- $*$

| lpar -- $($

| rpar -- $)$

inductive NT : Type

| E | S

abbrev G : CFG Sigma_RE :=

{ NT := NT

S := E

P :=

/-

E -> S | ε | $\emptyset$ | E $\cdot$ E | E+E | E* | (E)

S -> a | b | c

-/

{ (E, [inl S]),

(E, [inr epsilon]),

(E, [inr empty]),

(E, [inl E, inr dot,inl E]),

(E, [inl E, inr plus,inl E]),

(E, [inl E, inr star]),

(E, [inr lpar, inl E,inr rpar]),

(S, [inr a]),

(S, [inr b]),

(S, [inr c]) }

}

14. Show that

-- a $\cdot$ (b + c)

[a,dot,lpar,b,plus,c,rpar,star] $\in$ L G

by giving a derivation.

(10 marks)

15. The grammar G is ambiguous. Explain why.

How could this be fixed reflecting the conventions on how to read regular expressions (i.e. * binds more then . which binds more than *) ?

(15 marks)

Question 4

Answer the following questions about the Chomsky hierarchy.

16. What are the levels of the Chomsky hierarchy? What are the associated automata and grammar formalisms?

(9 marks)

17. Give examples of languages which are on a certain level but not on the previous one?

(9 marks)

18. Where do the decidable languages fit in? Why do they not correspond to a proper level of the hierarchy?

(7 marks)